

Elementary, My Dear Classifier: Interpretable Hypothesis Exclusion with Conformal Anchors

Cesare Barbera^{†1,2} (✉), Fabiola Bertocchi^{†2}, and Andrea Passerini²

¹ University of Pisa, cesare.barbera@phd.unipi.it

² Univeristy of Trento, {name}.{surname}@unitn.it

Abstract. The deployment of machine learning models in high-stakes domains is increasingly sought after, driven by their impressive performance. However, reliable decision support still faces two key limitations. First, most systems assume full information, whereas in high-stakes settings often only partial observations are available and additional evidence can be acquired. Supporting users in this feature acquisition process is crucial yet overlooked. Second, when error costs are highly asymmetric, it may be more important to reliably exclude dangerous alternatives than to commit to a specific hypothesis. To address these limitations, we propose *Conformal Anchors*, a model-agnostic framework that turns a downstream classifier into an interpretable hypothesis-rejection tool. Conformal Anchors answer questions of the form: which additional *features* should be observed, and which *values* should they take, to *reliably reject* a queried hypothesis? To this end, we identify simple, actionable and locally consistent feature-based rules that exclude the query from the conformal prediction set. We then show the conditions to obtain marginal guarantees on the erroneous exclusion of the true label. Our evaluation shows that Conformal Anchors achieve high precision, exhibit low empirical erroneous-exclusion rates and provide flexible trade-offs between coverage, compactness, and runtime via different construction strategies.

Keywords: XAI · Conformal Predictions · Hybrid-Decision Making.

1 INTRODUCTION

Recent advances in machine learning and artificial intelligence have led to models with remarkable predictive performance across a wide range of tasks, in some cases matching or surpassing human experts in specific domains [40, 22, 17, 33, 30]. However, in high-stakes settings, predictive accuracy alone is not enough [42]: a model may be accurate on average while still being unreliable for under-represented instances [44, 28, 9, 8] or sub-groups [23, 26, 7, 25, 41]. Moreover, different errors often have asymmetric consequences [16], *e.g.* failing to detect a severe disease is not equivalent to temporarily keeping an additional diagnosis under consideration. These reflections motivate local and exclusion-oriented decision support: rather than only recommending a single label, the system should provide instance-specific, uncertainty-aware and interpretable evidence about safe exclusion of a dangerous hypothesis.

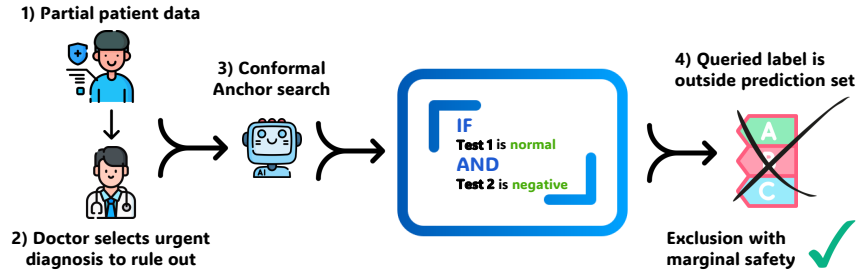


Fig. 1: Overview of the Conformal Anchors workflow in a medical setting. The method returns an interpretable rule over observed or acquirable features that excludes a queried diagnosis from the conformal prediction set.

Furthermore, high-stakes decision-making rarely occurs under full information: only partial evidence may be available initially, while additional information can often be acquired [3, 24]. The system should therefore support users throughout the acquisition process, but recommending what to observe next is not enough. In interactive decision support, the system should also explain which outcomes would matter and provide certified decision rules linking the acquired evidence to the conclusion it supports.

Example. A patient is rushed to the hospital and presents fever, cough, chest pain, and shortness of breath. These symptoms are compatible with several mild conditions, but also with a high-risk hypothesis such as pulmonary embolism, which would require a very urgent treatment. The physician may not yet need to commit to a final diagnosis, but must determine whether this urgent hypothesis can be safely ruled out. The relevant decision-support questions are therefore: which exams should be performed? Which outcomes would allow the physician to reject the urgent hypothesis? Can this rejection be made reliably?

We answer these questions with *Conformal Anchors* (Fig. 1), a framework that combines conformal prediction [43, 2] with Anchor explanations [35]. Conformal prediction identifies the set of labels that remain plausible for an instance while providing marginal coverage guarantees. Anchor explanations, in contrast, identify local and interpretable feature-value conditions under which a model behavior is stable. Conformal Anchors adapt this rule-construction mechanism to hypothesis rejection under partial information: they search over both observed and acquirable features to identify conditions under which a queried label is excluded from the conformal prediction set. As a result, exclusion from the set provides the basis for marginally-safe recommendations under the conditions we characterize, while Anchors make these recommendations interpretable.

Our Contributions. To summarize, our main contributions are:

- (i) we introduce *Conformal Anchors*, a framework for constructing locally consistent and actionable rules that support hypothesis exclusion (Section 4);
- (ii) we address the partial-information setting imputing missing values sampling from neighborhoods of the observed instance (Section 4.1);
- (iii) we develop practical construction strategies to find single or multiple Conformal Anchors, enabling diverse exclusion paths (Section 4.2);
- (iv) we identify the validity issues that arise when combining conformal prediction with anchor search, and characterize sufficient conditions under which Conformal Anchors control erroneous rejections (Section 4.3);
- (v) we evaluate a synthetic medical diagnosis scenario, analyzing desiderata of Conformal Anchors including reliability, rule quality, and runtime (Section 5).

2 RELATED WORK

This work lies at the intersection of conformal prediction, explainable artificial intelligence, and feature acquisition.

Conformal prediction. Conformal prediction [43, 2] is a model-agnostic approach to uncertainty quantification with finite-sample, distribution-free guarantees. Conformal methods return prediction sets that contain the true label with a user-specified probability [11, 31, 32]. Recent work has explored their potential to foster human-AI complementarity [13, 39, 38, 5, 12, 45, 15]. However, conformal sets indicate which labels remain plausible, but not which evidence justifies including or excluding a label. Recent works address this issue with counterfactual explanations for the conformal prediction set [1, 10, 29]. Conformal Anchors can operate under partial information and return locally stable feature-value rules, rather than pointwise counterfactual perturbations, to explain when a queried hypothesis can be excluded from the conformal prediction set.

Explainable Artificial Intelligence. XAI aims to make the behavior of complex models understandable to humans [20, 4]. Among XAI methods, rule-based explainers describe local model behavior through human-readable conditions [19, 34]. Anchor explanations [35] generate high-precision if-then rules that identify sufficient conditions under which a model prediction remains stable. Subsequent work improves and further investigates such explanations [14, 27]. Conformal Anchors build on this perspective, but they instead explain when a hypothesis is consistently excluded from a conformal prediction set.

Feature acquisition. Many high-stakes applications require decision-makers not only to make a prediction, but also to identify which information is needed before acting. Classical feature-acquisition methods select information to improve model performance, often under acquisition costs [36, 24, 3], with recent variants using explanations to guide acquisition [21]. Conformal Anchors not only identify which evidence should be acquired but also which values matter, and when those values certify the exclusion of a queried hypothesis.

3 BACKGROUND

Classification setting. Let $\mathbf{x} = (x_1, \dots, x_m) \in \mathcal{X}$ denote a feature vector and let $\mathcal{Y} = \{1, \dots, |\mathcal{Y}|\}$ be the label space. We consider a partial-information decision-support setting. At test time, the decision-maker initially observes only a fixed subset of features indexed by $I \subseteq \{1, \dots, m\}$, denoted $\mathbf{x}_I$. The remaining features $U = \{1, \dots, m\} \setminus I$ are initially unobserved, but can be acquired. Given $S \subset U$ selected features, the observed set is $O = I \cup S$, and the decision-maker sees $\mathbf{x}_O$. We assume access to a probabilistic classifier $f : \mathcal{X} \rightarrow \Delta^{|\mathcal{Y}|}$, mapping inputs to the probability simplex, and to $\hat{f}(\mathbf{x})$, its predictions. We do not require training records to contain full information, yet we refer to them as *fully realized*. In practice, this translates into allowing $I \subseteq O^{\text{train}}$ for training instances. This mirrors a medical setting where test-time patients present with only initial symptoms but the classifier is trained on historical records containing diagnoses, initial symptoms and the subset of exams ordered for each patient.

Conformal prediction. Given a trained classifier and a calibration set D^{cal} , conformal prediction constructs a set-valued predictor $\mathcal{C}_\alpha : \mathcal{X} \rightarrow 2^{\mathcal{Y}}$, where $\mathcal{C}_\alpha(\mathbf{x})$ is the set of labels considered plausible for instance $\mathbf{x}$ at miscoverage level $\alpha \in [0, 1]$. The central guarantee is marginal coverage: $\mathbb{P}(Y \in \mathcal{C}_\alpha(\mathbf{X})) \geq 1 - \alpha$ [6, 37], where $(\mathbf{X}, Y)$ denotes a fresh test example drawn exchangeably with D^{cal} . The elements of the conformal prediction set correspond to the hypotheses that cannot be marginally-safely excluded. Conversely, many decision-support systems interpret $y \notin \mathcal{C}_\alpha(\mathbf{x})$, as evidence for ruling out class y .

Anchor explanations. An anchor is a conjunction of interpretable feature predicates describing the prediction of a black-box classifier [35]. Let $A(\mathbf{x}) \in \{0, 1\}$ indicate whether rule A applies to instance $\mathbf{x}$, and let $\mathcal{D}$ be a perturbation distribution around $\mathbf{x}$. A rule on the features is an anchor if it applies to $\mathbf{x}$ and preserves the model prediction with high probability for perturbations satisfying the rule $\mathbf{z} \sim \mathcal{D}_A$. Its *precision* is $\mathbb{E}_{\mathcal{D}_A} [\mathbf{1}\{\hat{f}(\mathbf{z}) = \hat{f}(\mathbf{x})\}]$, while its *coverage* is $\mathbb{E}_{\mathcal{D}} [A(\mathbf{z})]$. Construction enforces precision above a fixed τ with probability at least $1 - \delta$, while maximizing coverage. The search typically leverages a beam procedure: candidate rules are expanded one predicate at a time, and only the B most promising partial rules are retained at each search depth. Larger beam sizes B therefore improve the breadth of the search at the expense of runtime.

4 CONFORMAL ANCHORS

Given a fully observed instance $\mathbf{x}$ and a queried hypothesis $y \in \mathcal{Y}$, we aim to identify interpretable conditions under which y can be rejected from the set of plausible labels. We can achieve this by directly modifying the original Anchors formulation. Formally, a *Conformal Anchor* for instance $\mathbf{x}$ and queried label y , is a conjunction of interpretable feature predicates A such that:

$$\mathbb{E}_{\mathbf{z} \sim \mathcal{D}_A} [\mathbf{1}\{y \notin \mathcal{C}_\alpha(\mathbf{z})\}] \geq \tau, \quad A(\mathbf{x}) = 1,$$

where $\tau \in (0, 1)$ is a user-specified precision threshold. Thus, A is a sufficient local condition under which, for most instances satisfying the rule, the queried hypothesis y is excluded from the conformal prediction set. Such a rule can be interpreted as a local, high-precision rejection certificate for y .

The conformal-anchor construction problem follows the same precision–coverage trade-off as standard anchors:

$$\max_A \text{Cov}(A) \quad \text{s.t.} \quad \mathbb{P}(\text{Prec}_\alpha(A, y) \geq \tau) \geq 1 - \delta,$$

where $\delta \in (0, 1)$ controls the confidence level. The key difference is the definition of precision: instead of measuring whether the point prediction is preserved, $\text{Prec}_\alpha(A, y)$ measures whether the queried hypothesis is excluded with high probability from the conformal prediction set when the rule applies.

However, Conformal Anchors introduce two critical challenges. First, under partial information, a useful rejection rule may depend on features whose values are not yet observed for the target instance. As a result, the method must reason not only about currently valid rules, but also about which additional unknown feature values would support the decision. Second, the anchor search can interfere with the exchangeability conditions underlying marginal conformal guarantees, potentially invalidating them. We address these two challenges next.

4.1 Conformal Anchors under Partial Information

Under partial information, the feature values that would exclude a candidate hypothesis y may not yet be observed for the target instance. Therefore, unlike standard anchors, we cannot generate rules solely from the currently observed features of $\mathbf{x}$. To address this, we look for local evidence: among instances that are similar to $\mathbf{x}$ on the initially observed features, what do their fully realized counterparts look like, and which of them provide evidence for excluding y ?

Formally, let $\mathcal{N}(\mathbf{x})$ be the set of most similar fully realized instances from a held-out anchor pool D^{anc} , where similarity is computed only on the initially observed features $\mathbf{x}_I$. For a hypothesis y , we define the exclusion neighborhood:

$$\mathcal{N}_y(\mathbf{x}) = \{\mathbf{x}' \in \mathcal{N}(\mathbf{x}) : y \notin C_\alpha(\mathbf{x}')\}.$$

This set contains locally similar and *fully realized* instances whose conformal prediction sets exclude y . It therefore represents the local empirical support on which exclusion rules for y should be evaluated. Denoting the perturbation distributions constrained to a neighborhood of $\mathbf{x}$ with $\mathcal{D}^\mathcal{N}$ and $\mathcal{D}_A^\mathcal{N}$, we define local precision and coverage measures as:

$$\text{Prec}_\alpha(A, y, \mathbf{x}) = \mathbb{E}_{\mathbf{z} \sim \mathcal{D}_A^\mathcal{N}} [\mathbf{1}\{y \notin C_\alpha(\mathbf{z})\}], \quad \text{Cov}(A, \mathbf{x}) = \mathbb{E}_{\mathbf{z} \sim \mathcal{D}^\mathcal{N}} [A(\mathbf{z})]$$

The use of local statistics for anchors is not new [14], but it admits a natural interpretation under partial information. Rather than seeking rules that fix model behavior regardless of unobserved features, we ask which observed and acquirable values anchor the exclusion of a hypothesis y across coherent completions. Thus, locality is both the regime in which rules are evaluated and the mechanism through which Conformal Anchors are constructed, as we show next.

4.2 Constructing Conformal Anchors

We propose practical construction strategies for Conformal Anchors. We first present the *one-shot mode*, which is based on a single reference sample and is the default procedure under full information. Then, for the partial-information setting, we introduce a *medoid-mode* which captures more diverse exclusion paths.

One-shot mode. The one-shot mode computes Conformal Anchors from a single reference instance. Under full information, this is $\mathbf{x}$ itself. This mode readily extends to partial-information settings where the exclusion neighborhood $\mathcal{N}_y(\mathbf{x})$ is also used to construct candidate rules. We sample a random reference instance from $\mathcal{N}_y(\mathbf{x})$, replace its initially observed block with $\mathbf{x}_I$, and use the resulting *fully realized* instance to define the predicate search space. However, a random selection can capture locally valid but not exhaustive exclusion patterns in heterogeneous neighborhoods. This limits the ability of one-shot mode to reflect multiple locally plausible exclusion paths.

Medoid mode. In medoid mode, we select multiple representative instances of $\mathcal{N}_y(\mathbf{x})$ via K -medoids clustering. We use Gower distance [18] to handle mixed feature types and missing values: it computes feature-wise normalized dissimilarities only on observed entries. Since each medoid is a real instance, its initial symptom block is replaced with $\mathbf{x}_I$. These constructions are then used separately to run an independent anchor search. The output is a pool of certified anchors, possibly capturing different local patterns that exclude the same hypothesis.

Since anchors constructed from different medoids may overlap, we apply a *union-pruning* step. Given returned anchors $\mathcal{A}$, we define *union coverage* as:

$$\text{Cov}(\mathcal{A}, \mathbf{x}) = \mathbb{E}_{\mathbf{z} \sim \mathcal{D}^{\mathcal{N}}} \left[\max_{A \in \mathcal{A}} A(\mathbf{z}) \right].$$

We consider two *union pruning* variants: *maximum-gain* (MG) (Algorithm 1), which selects the anchor with the largest marginal union-coverage gain, and *coverage-greedy* (CG) (Algorithm 2), which selects anchors by individual coverage. Both strategies stop when the increase in union coverage falls below a user-specified threshold ε or the set reaches a max size C , as per Corollary 1.

Overall, medoid mode can capture heterogeneous exclusion patterns, trading additional computational cost for greater diversity and expressivity. We conclude the section by analyzing the conformal validity of the anchor-search procedure and by characterizing sufficient conditions for controlling erroneous exclusions.

4.3 Marginal Guarantees

Our procedure estimates the precision of candidate rules evaluating whether the queried label is excluded from the conformal set over samples generated by the anchor perturbation mechanism. However, the split-conformal threshold computed on the calibration set D^{cal} is valid for examples exchangeable with D^{cal} .

Algorithm 1 MG pruning	Algorithm 2 CG pruning
1: $A_0 \in \arg \max_{A \in \mathcal{A}} \text{Cov}(A, \mathbf{x})$ 2: $\hat{\mathcal{A}} \leftarrow \{A_0\}$ 3: while $\mathcal{A} \setminus \hat{\mathcal{A}} \neq \emptyset$ do 4: $A^* \in \arg \max_{A \in \mathcal{A} \setminus \hat{\mathcal{A}}} \text{Cov}(\hat{\mathcal{A}} \cup \{A\}, \mathbf{x})$ 5: $\Delta \leftarrow \text{Cov}(\hat{\mathcal{A}} \cup \{A^*\}, \mathbf{x}) - \text{Cov}(\hat{\mathcal{A}}, \mathbf{x})$ 6: if $\Delta < \varepsilon \vee \hat{\mathcal{A}} = C$ then break 7: end if 8: $\hat{\mathcal{A}} \leftarrow \hat{\mathcal{A}} \cup \{A^*\}$ 9: end while 10: return $\hat{\mathcal{A}}$	1: $A_0 \in \arg \max_{A \in \mathcal{A}} \text{Cov}(A, \mathbf{x})$ 2: $\hat{\mathcal{A}} \leftarrow \{A_0\}$ 3: while $\mathcal{A} \setminus \hat{\mathcal{A}} \neq \emptyset$ do 4: $A^* \in \arg \max_{A \in \mathcal{A} \setminus \hat{\mathcal{A}}} \text{Cov}(A, \mathbf{x})$ 5: $\Delta \leftarrow \text{Cov}(\hat{\mathcal{A}} \cup \{A^*\}, \mathbf{x}) - \text{Cov}(\hat{\mathcal{A}}, \mathbf{x})$ 6: if $\Delta < \varepsilon \vee \hat{\mathcal{A}} = C$ then break 7: end if 8: $\hat{\mathcal{A}} \leftarrow \hat{\mathcal{A}} \cup \{A^*\}$ 9: end while 10: return $\hat{\mathcal{A}}$

Fig. 2: Maximum-Gain (left) and Coverage-greedy (right) pruning algorithms.

For an instance $\mathbf{x}$, perturbations $\mathbf{z} \sim \mathcal{D}_A$ may not satisfy exchangeability, since the rule A is imposed by construction. Thus, the set $C_\alpha(\mathbf{Z})$ does not automatically yield the desired $1 - \alpha$ marginal guarantee to contain Y . In what follows, we derive sufficient conditions under which the ordinary split-conformal threshold still yields a marginal guarantee, and we characterize the resulting bound.

To this end, consider a hypothetical perturbation-calibrated threshold $\tilde{q}_\rho$ ³, defined so that, with probability at least $1 - \alpha$, every candidate anchor has at least a ρ -fraction of perturbations whose conformal sets contain the true label. We term this condition *candidate-wise perturbation validity* (C1) (Appendix B.1). Notably, when $\hat{q} \geq \tilde{q}_\rho$ (*threshold dominance*), the perturbation-calibrated sets are included in the conformal sets induced by $\hat{q}$, so conformal sets constructed for the perturbations using $\hat{q}$ inherit the $1 - \alpha$ guarantee. We term this condition *safety under threshold dominance* (C2) (Appendix B.2).

We now present our main result: we identify when the ordinary split-conformal threshold can be used to obtain a marginal guarantee and characterize the resulting bound. The assumptions are stated in Appendix A. They concern (A1) *ordinary* and (A2) *candidate-wise perturbation exchangeability*, (A3) *the auxiliary perturbation-label distribution* used only for the analysis, (A4) *precision certification*, (A5) *precision-quantile compatibility*, (A6) *population-level separation*, and (A7) *concentration of the empirical CDFs*. Under A(1)–A(7), we derive a *finite-sample bound* (C3, Appendix B.3) such that *threshold dominance* fails with probability at most η . Therefore, under the sufficient conditions (C1)–(C3) the ordinary split-conformal threshold yields the marginal guarantee in Theorem 1. Notably, the guarantee decomposes erroneous exclusions into three interpretable sources of failure: ordinary conformal miscoverage (α), failure of the anchor precision certificate (δ), and failure of threshold dominance (η). Denoting

³ $\tilde{q}_\rho$ is only an analytical object (construction details in Appendix B). The implemented method uses the ordinary split-conformal threshold $\hat{q}$.

by $A_Y(\mathbf{X}_O) = 1$ the event that a returned anchor excluding the true label Y applies to $\mathbf{X}_O$, we obtain the following probabilistic bound.

Theorem 1 (Informal Conformal Anchors guarantee). *Assume (A1) – (A7) and let the threshold-dominance event fail with probability at most η (Lemma 3 derived in Appendix B.3). Using the original split-conformal-threshold $\hat{q}$ yields:*

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1) \leq \alpha + \delta + \eta.$$

Proof. Proof is provided in Appendix B.

Theorem 1 should be read as a sufficient-condition guarantee and, more importantly, as a characterization of why high-precision exclusion can be compatible with marginal control, rather than as an unconditional conformal-validity statement for arbitrary perturbations. The key observation is that the exclusion objective changes the relevant calibration requirement. Since Conformal Anchors certify high-precision rejection rules, erroneous exclusions correspond to rules that exclude the true label on a large fraction of their perturbations. This shifts the relevant safety condition to the lower tail of the perturbation score distribution: since a certified anchor excludes the queried label on at least a τ -fraction of perturbations, the relevant calibration event is that, with probability at least $1 - \alpha$, a ρ -fraction of perturbations retain the true label for $\rho > 1 - \tau$. Thus, the probability of erroneous exclusion decomposes into the three sources of failure: ordinary conformal misscoverage (α), failure of the anchor precision certificate (δ), and failure of the low-quantile dominance condition (η).

We conclude the section with a corollary controlling erroneous rejections when multiple anchors are returned for the same hypothesis.

Corollary 1 (Budgeted control for multiple returned anchors). *Under the assumptions of Theorem 1, assume to return at most C Conformal Anchors for any label. If the guarantee of Theorem 1 holds for each object with budgets*

$$\alpha' := \frac{\alpha}{C}, \quad \delta' := \frac{\delta}{C}, \quad \eta' := \frac{\eta}{C},$$

then

$$\mathbb{P}\left(\exists c \in \{1, \dots, C\} : A_Y^{(c)}(\mathbf{X}_O) = 1\right) \leq \alpha + \delta + \eta.$$

Proof. Proof is provided in Appendix B.5.

This corollary extends the marginal guarantee to the case of multiple returned Conformal Anchors, allowing, under the same sufficient conditions of Theorem 1, marginally safe exclusion when evaluating more exhaustive exclusion patterns.

5 EXPERIMENTAL EVALUATION

In this section, we evaluate the following properties of Conformal Anchors:

- **P1: Reliability** *i.e.* are erroneous exclusions rare on held-out data?
- **P2: Rule quality**, *i.e.* are exclusion conditions stable and broadly applicable?
- **P3: Complexity**, *i.e.* what is the trade-off between expressivity and runtime?

Code is available at: <https://github.com/fabertocchi/conformal-anchors>

Table 1: Simplified training and test patient records and Conformal Anchor example. The rule is returned for the test-time patient and each queried disease.

Feature	Patient 1 (Pneumonia)	Patient 2 (Gastritis)	Test-time patient (Peptic ulcer)	Conf. Anch. (Gastric cancer)	Conf. Anch. (Peptic ulcer)
<i>Symptoms</i>					
Fever severity	7	3	3	—	—
Cough severity	7	0	4	—	—
Chest pain severity	4	0	2	—	—
Abdominal pain	0	9	9	—	—
Nausea	2	3	7	—	—
<i>Diagnostic tests</i>					
FEV1% predicted	34.50	—	—	—	—
Chest X-ray score	9	—	—	—	—
WBC count	18k	—	—	—	—
Hemoglobin (g/dL)	—	15.05	—	> 6.50	—
Gastric pH	—	1.50	—	≤ 3.00	> 5.00

5.1 Experimental setup

Dataset. We evaluate Conformal Anchors in a synthetic medical diagnosis scenario. Each instance represents a patient described by symptoms and diagnostic tests, and the label corresponds to one disease among a set of diagnostic classes. We generate 100k instances and we report in Table 1 simplified samples of the records of training and test instances. We detail dataset construction in Appendix C and provide full random samples in Appendix C.3. For evaluation, we separate the data into four disjoint parts: (i) a training set (65%); (ii) a calibration set (10%); (iii) an anchor-search set (10%); and (iv) a held-out test set used exclusively for evaluation (15%). Then, to keep runtime manageable, we subsample 1000 test instances to evaluate Conformal Anchors.

Methods. We compare the *one-shot* (OS) mode, the *maximum-gain* (MG) mode and the *coverage-greedy* (CG) mode. For all modes we select an optimal beam-size of 5 (Appendix D.2) and neighborhood size is set to 100. The precision threshold and its error tolerance are $\tau = 0.95$ and $\delta = 0.01$, respectively. Conformal miscoverage is $\alpha = 0.01$, instead. We detail all hyperparameters and their selection criteria in Appendix D.

Statistics. For **P1**, we test empirically the rate of true-label exclusion among test instances. For **P2**, we report empirical precision, coverage and union-coverage. For **P3**, we compare running-times with number of anchors returned, number of unique features among them and number of features per anchor.

5.2 Experimental Results

P1: Conformal Anchors exhibit low erroneous-exclusion rates. Threshold dominance is a diagnostic assumption on the perturbation mechanism: the anchor-induced perturbations must not make the nonconformity scores substantially harder than ordinary calibration examples at the relevant low quantile.

Table 2: Empirical safety statistics over 1000 instances and two beam sizes B .

Mode	$B = 5$			$B = 1$		
	Erroneous exclusions	C	$(\alpha + \delta)C$	Erroneous exclusions	C	$(\alpha + \delta)C$
OS	0.016 ✓	10	0.2	0.004 ✓	1	0.02
MG	0.050 ✓	8	0.16	0.011 ✓	8	0.16
CG	0.016 ✓	6	0.12	0.010 ✓	7	0.14

Table 3: Precision, coverage, and union coverage for 1000 instances.

Mode	Precision	Coverage	Union coverage
OS	0.981 ± 0.026	0.361 ± 0.130	0.361 ± 0.141
MG	0.976 ± 0.020	0.392 ± 0.108	0.581 ± 0.095
CG	0.976 ± 0.020	0.392 ± 0.108	0.429 ± 0.117

Although the auxiliary threshold is not observed, this condition motivates using local perturbations and high-precision anchors, and it can be indirectly stress-tested by measuring the erroneous exclusions rate on held-out data (Table 2). This rate remains below the budget $(\alpha + \delta)C < (\alpha + \delta + \eta)C$ for all methods. Larger beam sizes increase the maximum number C of valid anchors found, but the corresponding erroneous-exclusion rates remain below the empirical budget.

P2: Conformal Anchors provide stable and broadly applicable rules.

Table 3 reports precision, coverage, and union coverage. The first two apply only to the main anchor, *i.e.* the valid anchor with highest coverage, while the latter to all returned objects. All methods similarly achieve high empirical precision (≈ 0.98): the returned anchors define stable exclusion conditions when they apply. Instead, MG and CG improve over OS in terms of coverage and yield very similar values (≈ 0.39) as they share the same anchor search space and find the same main anchors. The main differences arise in union coverage. Medoid-based strategies improve over OS, with MG achieving larger values (≈ 0.58) than the rest. This suggests that selecting representative medoids exposes complementary exclusion patterns within $\mathcal{N}_y(\mathbf{x})$. Overall, the returned anchors exhibit high rule quality, combining strong stability with substantial local coverage.

P3: Conformal Anchors provide flexible trade-offs between expressivity and computational cost. Table 4 compares explanation complexity and runtime across construction modes. OS returns the largest explanation sets, with ≈ 5 anchors per instance on average, whereas MG (≈ 4) and CG (≈ 2) return smaller, pruned sets. Individual anchors remain compact across all modes, a key requirement for interpretability, involving ≈ 2 features per rule. MG yields the highest number of unique features across anchors (≈ 5), while CG produces the

Table 4: Explanation complexity and runtime in seconds for 1000 instances.

Mode	# anchors	# features/anchor	# unique features	mean runtime
OS	4.60 ± 1.75	2.24 ± 0.84	3.38 ± 1.60	3.10 ± 5.92
MG	3.96 ± 1.23	2.27 ± 0.47	4.95 ± 1.73	15.04 ± 20.55
CG	1.53 ± 0.81	2.23 ± 0.75	2.68 ± 1.28	15.04 ± 20.55

most compact explanation sets with (≈ 3) unique features. Since MG directly maximizes gains in union coverage, it favors complementary anchors and therefore broader sets. Instead, CG prioritizes individual coverage, so selections may overlap, producing more compact but less diverse explanations. Runtimes differ substantially across modes: OS requires ≈ 3 seconds per instance on average, while medoid-based strategies require ≈ 15 seconds, although with large standard deviations.

Concluding remark. Overall, our analysis shows that all construction modes achieve high empirical precision and low erroneous-exclusion rates in this setting. However, they induce different trade-offs between runtime, compactness, and neighborhood coverage: OS favors efficiency, MG favors diverse exclusion paths and CG favors compact explanation sets with lower cognitive load.

6 CONCLUSIONS

Conclusions. We introduced Conformal Anchors, a framework for interpretable and probabilistically grounded hypothesis rejection in multiclass classification. We first formalized the framework and proposed practical construction strategies that adapt to the partial-information setting. Then, we derived sufficient conditions to obtain the Conformal Anchors’ marginal guarantees. Our proposals empirically exhibit erroneous-exclusion rates consistent with our sufficient-condition analysis (Section 4.3), achieve high precision and coverage and provide flexible trade-offs between expressivity and runtime.

Limitations and Future Work. In the high-precision regime ($\tau = 0.95$), threshold dominance is required only at a low perturbation quantile ($\rho > 0.05$). Still, the auxiliary threshold is not observable and marginal safety must be evaluated in practice. Moreover, the resulting budget is conservative and the empirical erroneous-exclusion rates are substantially below it. Tightening the multiple-anchor analysis remains an important direction for future work. We further assume equally costly features and single-hypothesis exclusions per instance. We plan to extend this work to a sequential, cost-sensitive feature-acquisition setting, where evidence is acquired adaptively and hypotheses are ruled out over time. Finally, our evaluation is restricted to a synthetic setting. Future work should validate Conformal Anchors on real-world data and via a user study in realistic decision-support workflows.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

- [1] Altmeyer, P., Farmanbar, M., van Deursen, A., Liem, C.C.: Faithful model explanations through energy-constrained conformal counterfactuals. In: Proceedings of the AAAI conference on artificial intelligence. vol. 38, pp. 10829–10837 (2024)
- [2] Angelopoulos, A.N., Bates, S.: A gentle introduction to conformal prediction and distribution-free uncertainty quantification. CoRR **abs/2107.07511** (2021)
- [3] Aronsson, L., Rahbar, A., Chehreghani, M.H.: A survey on active feature acquisition strategies. arXiv preprint arXiv:2502.11067 (2025)
- [4] Arrieta, A.B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., et al.: Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai. *Information fusion* **58**, 82–115 (2020)
- [5] Babbar, V., Bhatt, U., Weller, A.: On the utility of prediction sets in human-ai teams. arXiv preprint arXiv:2205.01411 (2022)
- [6] Balasubramanian, V., Ho, S.S., Vovk, V.: Conformal prediction for reliable machine learning: theory, adaptations and applications. Newnes (2014)
- [7] Baldeschi, R.C., Di Gregorio, S., Fioravanti, S., Fusco, F., Guy, I., Haimovich, D., Leonardi, S., Linder, F., Perini, L., Russo, M., et al.: Multicalibration yields better matchings. arXiv preprint arXiv:2511.11413 (2025)
- [8] Barbera, C., Perini, L., De Toni, G., Passerini, A., Pugnana, A.: Divide et calibra: Multiclass local calibration via vector quantization. arXiv preprint arXiv:2605.21060 (2026)
- [9] Barbera, C., Perini, L., Toni, G.D., Passerini, A., Pugnana, A.: Multiclass Local Calibration With the Jensen-Shannon Distance. In: AISTATS (2026)
- [10] Bilkhoo, A., Hosseini, M., Kazemi, M., Paoletti, N.: Confex: Uncertainty-aware counterfactual explanations with conformal guarantees. arXiv preprint arXiv:2510.19754 (2025)
- [11] Chzhen, E., Denis, C., Hebiri, M., Lorieul, T.: Set-valued classification—overview via a unified framework. arXiv preprint arXiv:2102.12318 (2021)
- [12] Cresswell, J.C., Sui, Y., Kumar, B., Vouitsis, N.: Conformal prediction sets improve human decision making. arXiv preprint arXiv:2401.13744 (2024)
- [13] De Toni, G., Okati, N., Thejaswi, S., Straitouri, E., Gomez-Rodriguez, M.: Towards human-ai complementarity with prediction sets. *Advances in Neural Information Processing Systems* **37**, 31380–31409 (2024)
- [14] Delaunay, J., Galárraga, L., Largouët, C.: Improving anchor-based explanations. In: Proceedings of the 29th ACM International Conference on Information & Knowledge Management. pp. 3269–3272 (2020)
- [15] Donahue, K., Gollapudi, S., Kollias, K.: When are two lists better than one?: Benefits and harms in joint decision-making. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 38, pp. 10030–10038 (2024)

- [16] Elkan, C.: The foundations of cost-sensitive learning. In: IJCAI. pp. 973–978. Morgan Kaufmann (2001)
- [17] Esteva, A., Kuprel, B., Novoa, R.A., Ko, J., Swetter, S.M., Blau, H.M., Thrun, S.: Dermatologist-level classification of skin cancer with deep neural networks. *nature* **542**(7639), 115–118 (2017)
- [18] Gower, J.C.: A general coefficient of similarity and some of its properties. *Biometrics* pp. 857–871 (1971)
- [19] Guidotti, R., Monreale, A., Ruggieri, S., Pedreschi, D., Turini, F., Giannotti, F.: Local rule-based explanations of black box decision systems. *arXiv preprint arXiv:1805.10820* (2018)
- [20] Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., Pedreschi, D.: A survey of methods for explaining black box models. *ACM computing surveys (CSUR)* **51**(5), 1–42 (2018)
- [21] Guney, O.B., Saichandran, K.S., Elzokm, K., Zhang, Z., Kolachalama, V.B.: Active feature acquisition via explainability-driven ranking. *Proceedings of machine learning research* **267**, 20748 (2025)
- [22] He, K., Zhang, X., Ren, S., Sun, J.: Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In: *Proceedings of the IEEE international conference on computer vision*. pp. 1026–1034 (2015)
- [23] Hébert-Johnson, U., Kim, M., Reingold, O., Rothblum, G.: Multicalibration: Calibration for the (computationally-identifiable) masses. In: *International Conference on Machine Learning*. pp. 1939–1948. PMLR (2018)
- [24] Ji, S., Carin, L.: Cost-sensitive feature acquisition and classification. *Pattern Recognition* **40**(5), 1474–1485 (2007)
- [25] Jin, H.H., Ding, Z., Ngo, D.D., Wu, Z.S.: Discretization-free multicalibration through loss minimization over tree ensembles. *arXiv preprint arXiv:2505.17435* (2025)
- [26] Jung, C., Lee, C., Pai, M., Roth, A., Vohra, R.: Moment multicalibration for uncertainty estimation. In: *Conference on Learning Theory*. pp. 2634–2678. PMLR (2021)
- [27] Lopardo, G., Precioso, F., Garreau, D.: Understanding post-hoc explainers: The case of anchors. *arXiv preprint arXiv:2303.08806* (2023)
- [28] Luo, R., Bhatnagar, A., Bai, Y., Zhao, S., Wang, H., Xiong, C., Savarese, S., Ermon, S., Schmerling, E., Pavone, M.: Local calibration: metrics and recalibration. In: *UAI. Proceedings of Machine Learning Research*, vol. 180, pp. 1286–1295. PMLR (2022)
- [29] Maalej, A., Sönströd, C., Johansson, U.: Counterfactual explanations for conformal prediction sets. *Proceedings of Machine Learning Research* **266**, 1–20 (2025)
- [30] Matek, C., Schwarz, S., Spiekermann, K., Marr, C.: Human-level recognition of blast cells in acute myeloid leukaemia with convolutional neural networks. *Nature Machine Intelligence* **1**(11), 538–544 (2019)
- [31] Mortier, T., Wydmuch, M., Dembczyński, K., Hüllermeier, E., Waegeman, W.: Efficient set-valued prediction in multi-class classification. *Data Mining and Knowledge Discovery* **35**(4), 1435–1469 (2021)

- [32] Nguyen, V.L., Hüllermeier, E.: Multilabel classification with partial abstention: Bayes-optimal prediction under label independence. *Journal of Artificial Intelligence Research* **72**, 613–665 (2021)
- [33] Rajpurkar, P.: Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning. *ArXiv abs/1711* **5225** (2017)
- [34] Ribeiro, M.T., Singh, S., Guestrin, C.: "why should i trust you?" explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. pp. 1135–1144 (2016)
- [35] Ribeiro, M.T., Singh, S., Guestrin, C.: Anchors: High-precision model-agnostic explanations. In: *AAAI*. pp. 1527–1535. AAAI Press (2018)
- [36] Saar-Tsechansky, M., Melville, P., Provost, F.: Active feature-value acquisition. *Management Science* **55**(4), 664–684 (2009)
- [37] Shafer, G., Vovk, V.: A tutorial on conformal prediction. *Journal of machine learning research* **9**(3) (2008)
- [38] Straitouri, E., Rodriguez, M.G.: Designing decision support systems using counterfactual prediction sets. *arXiv preprint arXiv:2306.03928* (2023)
- [39] Straitouri, E., Wang, L., Okati, N., Rodriguez, M.G.: Improving expert predictions with conformal prediction. In: *International Conference on Machine Learning*. pp. 32633–32653. PMLR (2023)
- [40] Taigman, Y., Yang, M., Ranzato, M., Wolf, L.: Deepface: Closing the gap to human-level performance in face verification. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. pp. 1701–1708 (2014)
- [41] Tax, N., Perini, L., Linder, F., Haimovich, D., Karamshuk, D., Okati, N., Vojnovic, M., Apostolopoulos, P.A.: MCGrad: Multicalibration at Web Scale. In: *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1 (KDD 2026)* (2026). <https://doi.org/10.1145/3770854.3783954>
- [42] Van Calster, B., McLernon, D.J., Van Smeden, M., Wynants, L., Steyerberg, E.W., diagnostic tests, T.G.E., prediction models' of the STRATOS initiative Bossuyt Patrick Collins Gary S. Macaskill Petra McLernon David J. Moons Karel GM Steyerberg Ewout W. Van Calster Ben van Smeden Maarten Vickers Andrew J.: Calibration: the achilles heel of predictive analytics. *BMC medicine* **17**(1), 230 (2019)
- [43] Vovk, V., Gammernan, A., Shafer, G.: *Algorithmic learning in a random world*. Springer (2005)
- [44] Xiong, M., Deng, A., Koh, P.W.W., Wu, J., Li, S., Xu, J., Hooi, B.: Proximity-informed calibration for deep neural networks. In: *NeurIPS* (2023)
- [45] Zhang, D., Chatzimpampas, A., Kamali, N., Hullman, J.: Evaluating the utility of conformal prediction sets for ai-advised image labeling. In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. pp. 1–19 (2024)

Appendix

A Assumptions

We state the assumptions used in the marginal guarantee analysis.

Let $D^{\text{cal}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ denote the split-conformal calibration set, and let $s(\mathbf{x}, y)$ be the nonconformity score used to construct conformal prediction sets.

(A1). *Exchangeability of ordinary calibration scores.* The ordinary calibration scores

$$S_i = s(\mathbf{x}_i, y_i), \quad i = 1, \dots, n,$$

are exchangeable with the score of a fresh test example

$$S_{\text{test}} = s(\mathbf{X}, Y).$$

This is the standard exchangeability assumption required for split conformal prediction.

(A2). *Candidate-wise perturbation exchangeability.* For each calibration instance $\mathbf{x}_i$, we run the same anchor-search procedure used at test time. Let $\mathcal{A}_i^{\text{pool}}$ be the set of candidate anchors evaluated for $\mathbf{x}_i$. For each $A \in \mathcal{A}_i^{\text{pool}}$, let

$$\mathcal{Z}_i^A = \{(\mathbf{z}_{i,1}^A, y_{i,1}^A), \dots, (\mathbf{z}_{i,\tilde{n}_i^A}^A, y_{i,\tilde{n}_i^A}^A)\}$$

be the perturbation bundle generated while evaluating A . We assume that the same perturbation mechanism is applied to calibration and test instances, and that the candidate-wise aggregated scores

$$S_i^\rho = \max_{A \in \mathcal{A}_i^{\text{pool}}} \text{Quantile}_\rho \{s(\mathbf{z}_{i,j}^A, y_{i,j}^A) : j = 1, \dots, \tilde{n}_i^A\}$$

are exchangeable with the corresponding test score

$$S_{\text{test}}^\rho = \max_{A \in \mathcal{A}_{\text{test}}} \text{Quantile}_\rho \{s(\mathbf{z}_{\text{test},j}^A, y_{\text{test},j}^A) : j = 1, \dots, \tilde{n}_{\text{test}}^A\}.$$

This assumption allows split conformal prediction to be applied to the candidate-wise perturbation scores.

(A3). *Auxiliary perturbation-label distribution.* The labels $y_{i,j}^A$ associated with perturbed samples are understood with respect to an auxiliary perturbation-label distribution. This distribution is used only for the analysis, in order to define the perturbation-calibrated threshold $\tilde{q}_\rho$. The implemented method does not require access to labeled synthetic perturbations when it uses the ordinary split-conformal threshold $\hat{q}$.

(A4). *Precision certification.* For every queried label y , let A_y be the Conformal Anchor returned for excluding y . The precision certification procedure satisfies

$$\mathbb{P}(\text{Prec}(A_y(\mathbf{X}_O)) \geq \tau) \geq 1 - \delta.$$

(A5). *Precision-quantile compatibility.* The target precision τ and the perturbation quantile level ρ satisfy

$$\tau > 1 - \rho.$$

This condition ensures that a certified τ -precision anchor for the true label would exclude the true label on a larger fraction of perturbations than allowed by the ρ -quantile perturbation-calibration event.

(A6). *Population-level separation conditions.* Let F denote the CDF of the ordinary calibration scores and G_ρ the CDF of the auxiliary aggregated perturbation scores. Define $\zeta_\alpha := \min\{n, \lceil (n+1)(1-\alpha) \rceil\}$. We assume that there exist $t \in \mathbb{R}$ and margins $\gamma_F, \gamma_G > 0$ such that

$$F(t) \leq \frac{\zeta_\alpha - 1}{n} - \gamma_F, \quad G_\rho(t) \geq \frac{\zeta_\alpha}{n} + \gamma_G.$$

(A7). *Concentration of empirical CDFs.* The empirical CDFs of the ordinary calibration scores and of the auxiliary aggregated perturbation scores satisfy Hoeffding-type concentration bounds at the separating threshold t .

B Proofs

In what follows, we derive the marginal guarantees stated in Theorem 1 and Corollary 1. The proof is organized around three lemmas that isolate the main ingredients of the argument: candidate-wise perturbation validity, safety under threshold dominance, and a finite-sample bound for the dominance event. Combining these ingredients yields the final marginal guarantee, and a budgeted union bound extends the result to multiple returned anchors.

We begin characterizing the perturbation-level calibration requirement introducing an hypothetical perturbed calibration set $\tilde{D}^{\text{cal}}$. Each instance $(\mathbf{x}_i, y_i) \in D^{\text{cal}}$ is treated as a pseudo-test instance and we assume to observe the test-time anchor-search procedure. The procedure may consider multiple candidate anchors $A \in \mathcal{A}_i^{\text{pool}}$ and all the perturbations generated for the candidate are collected into a bundle $\mathcal{Z}_i^A = \{(\mathbf{z}_{i,1}^A, y_{i,1}^A), \dots, (\mathbf{z}_{i,\tilde{n}_i^A}^A, y_{i,\tilde{n}_i^A}^A)\}$. Here $y_{i,j}^A$ denotes the label that would be associated with the perturbed sample under the auxiliary perturbation-label distribution. This object is also hypothetical and used only for the analysis. The elements of $\mathcal{Z}_i^A$ are not **i.i.d.**, since they share the values of A . Therefore perturbation scores are aggregated at the level of the original instance-anchor pair. We therefore assume to compute the conservative aggregated score:

$$S_i^\rho = \max_{A \in \mathcal{A}_i^{\text{pool}}} \text{Quantile}_\rho \{s(\mathbf{z}_{i,j}^A, y_{i,j}^A) : j = 1, \dots, \tilde{n}_i^A\}$$

The perturbed conformal threshold is computed from the aggregated scores as:

$$\tilde{\rho} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n} (S_1^\rho, \dots, S_n^\rho), \quad n = |D^{\text{cal}}|$$

The following lemma illustrates the resulting marginal guarantee.

B.1 Proof of Lemma 1.

Lemma 1 (Candidate-wise perturbation marginal validity). *Assume (A1)-(A3). With probability at least $1 - \alpha$, for any candidate A , at least a ρ -fraction of the perturbations contain the true label in their conformal set.*

$$\mathbb{P}(S_{\text{test}}^\rho \leq \tilde{q}_\rho) \geq 1 - \alpha$$

Proof. The result is an immediate application of split conformal prediction to the aggregated scores S_i^ρ . Under assumption (A2), the standard conformal quantile choice gives $\mathbb{P}(S_{\text{test}}^\rho \leq \tilde{q}_\rho) \geq 1 - \alpha$. Since S_{test}^ρ is the maximum among evaluated candidates of the ρ -quantiles of the perturbation scores, the event $S_{\text{test}}^\rho \leq \tilde{q}_\rho$ means that simultaneously for every $A \in \mathcal{A}_{\text{test}}$ at least a ρ -fraction of the generated perturbations have nonconformity score below the calibrated threshold. Hence contain the true label in the corresponding conformal prediction set.

q.e.d.

We further assume access to a split-conformal threshold on samples in D^{cal} :

$$\hat{q} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n} (s(\mathbf{x}_1, y_1), \dots, s(\mathbf{x}_n, y_n))$$

and $q_{\max} = \max\{\hat{q}, \tilde{q}_\rho\}$ defines the hypothetical conservative threshold for Conformal Anchors and the respective conformal set is $\mathcal{C}_\alpha(\mathbf{x}) = \{y : s(\mathbf{x}, y) \leq q_{\max}\}$.

With this threshold, both the perturbations and original data distributions are conservatively calibrated. Problematically, calibrating the perturbation distribution in practice would require labeled examples, but perturbations are synthetic and unlabeled. In what follows, we use $\tilde{q}_\rho$ to characterize a sufficient condition under which the ordinary split-conformal threshold $\hat{q}$ is already conservative enough. We start by deriving marginal guarantees on the erroneous exclusion of the true label when $q_{\max} = \hat{q}$ as the following lemma illustrates.

B.2 Proof of Lemma 2.

Lemma 2 (From perturbation calibration to anchor safety). *Assume (A1)-(A5) and for any label y , let A_y denote the Conformal Anchor returned to exclude y . If $\hat{q} \geq \tilde{q}$, the probability of returning a Conformal Anchor that applies to X_O and excludes the true label is bounded as follows:*

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1) \leq \alpha + \delta.$$

Proof. Let

$$\mathcal{E}_Y(\mathbf{X}_O) = \{\text{Prec}(A_Y(\mathbf{X}_O)) \geq \tau\}$$

be the event that the returned anchor for the true label Y satisfies the required precision constraint. By assumption (A4),

$$\mathbb{P}(\mathcal{E}_Y(\mathbf{X}_O)^c) \leq \delta.$$

We decompose

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1) = \mathbb{P}(A_Y(\mathbf{X}_O) = 1, \mathcal{E}_Y(\mathbf{X}_O)) + \mathbb{P}(A_Y(\mathbf{X}_O) = 1, \mathcal{E}_Y(\mathbf{X}_O)^c).$$

The second term is bounded by

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1, \mathcal{E}_Y(\mathbf{X}_O)^c) \leq \mathbb{P}(\mathcal{E}_Y(\mathbf{X}_O)^c) \leq \delta.$$

By Lemma 1 and the definition of the max-aggregated perturbation score,

$$\mathbb{P}(\exists A \in \mathcal{A}_{\text{test}} : \text{Excl}_Y(A, X_O) > 1 - \rho) \leq \alpha$$

where

$$\text{Excl}_Y(A, \mathbf{X}_O) = \frac{1}{\tilde{n}^A} \sum_{j=1}^{\tilde{n}^A} \mathbf{1} \left\{ Y \notin \mathcal{C}_\alpha(\tilde{\mathbf{Z}}_j^A) \right\}$$

since the returned anchor A_Y belongs to $\mathcal{A}_{\text{test}}$, we have

$$\mathbb{P}(\text{Excl}_Y(A_Y, X_O) > 1 - \rho) \leq \alpha.$$

Because the perturbations are generated by enforcing the returned anchor, the anchor applies to every evaluated perturbation. Hence, on the event $\mathcal{E}_Y(\mathbf{X}_O)$, if an anchor for excluding the true label Y is returned, then the true label is excluded from at least a τ -fraction of the generated perturbations:

$$\text{Excl}_Y(A_Y, \mathbf{X}_O) \geq \tau.$$

By (A5), $\tau > 1 - \rho$ therefore:

$$\{A_Y(\mathbf{X}_O) = 1, \mathcal{E}_Y(\mathbf{X}_O)\} \subseteq \{\text{Excl}_Y(A_Y, \mathbf{X}_O) > 1 - \rho\}.$$

and,

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1, \mathcal{E}_Y(\mathbf{X}_O)) \leq \mathbb{P}(\text{Excl}_Y(A_Y, \mathbf{X}_O) > 1 - \rho) \leq \alpha.$$

Combining again with the bad certification event gives

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1) \leq \alpha + \delta.$$

q.e.d.

This result assumes $\hat{q} > \tilde{q}$ and shows that the key condition for marginal guarantees against erroneous label exclusions is $\tau > 1 - \rho$. Since Conformal Anchors are used with high target precision τ , the required perturbation coverage level ρ can be small. Therefore, for marginal guarantees, validity over most anchor perturbations is not a requirement. It suffices that empirically the split-conformal threshold dominates the perturbation-calibrated threshold at some $\rho > 1 - \tau$. The next lemma gives a finite-sample bound for this dominance event.

B.3 Proof of Lemma 3.

Lemma 3 (A finite-sample dominance bound). *Assume (A1) – (A7). Let:*

$$\hat{q} := S_{(\zeta_\alpha)}, \quad \tilde{q}_\rho := S_{(\zeta_\alpha)}^\rho,$$

then:

$$\mathbb{P}(\tilde{q}_\rho \leq \hat{q}) \geq 1 - \underbrace{(\exp(-2n\gamma_F^2) + \exp(-2n\gamma_G^2))}_\eta.$$

Proof. Let $\hat{F}_n$ and $\hat{G}_n$ be the empirical CDFs of the ordinary scores and auxiliary perturbation scores, respectively. By the assumed Hoeffding-type concentration bounds,

$$\mathbb{P}\left(\hat{F}_n(t) \leq \frac{\zeta_\alpha - 1}{n}\right) \geq 1 - \exp(-2n\gamma_F^2),$$

because $F(t) \leq (\zeta_\alpha - 1)/n - \gamma_F$. Similarly,

$$\mathbb{P}\left(\hat{G}_n(t) \geq \frac{\zeta_\alpha}{n}\right) \geq 1 - \exp(-2n\gamma_G^2),$$

because $G_\rho(t) \geq \zeta_\alpha/n + \gamma_G$. By a union bound, both events hold with probability at least:

$$1 - \exp(-2n\gamma_F^2) - \exp(-2n\gamma_G^2).$$

On this event, $\hat{G}_n(t) \geq \zeta_\alpha/n$, so at least ζ_α auxiliary perturbation scores are at most t . Therefore,

$$\tilde{q}_\rho \leq t.$$

Likewise, $\hat{F}_n(t) \leq (\zeta_\alpha - 1)/n$, so at most $\zeta_\alpha - 1$ ordinary calibration scores are at most t . Therefore, the ζ_α -th ordinary order statistic satisfies

$$\hat{q} \geq t.$$

Hence

$$\tilde{q}_\rho \leq t \leq \hat{q},$$

and therefore

$$\tilde{q}_\rho \leq \hat{q}.$$

q.e.d.

In high-precision regimes, the separation condition concerns only a low perturbation score quantile. The assumption fails only when anchor-generated perturbations are so difficult for the classifier that even their low $(1 - \tau^-)$ -quantile⁴ nonconformity score lies above the ordinary ζ_α -th conformal calibration threshold on real data. In that case, the standard conformal threshold is not conservative enough. Notably, the locality of the perturbation mechanism mitigates this.

⁴ τ^- denotes a value infinitesimally smaller than precision τ .

Since perturbations are generated from neighbors rather than from arbitrary global samples, they remain close to the local empirical support.

We now introduce the final bound on the erroneous exclusion of the true label with Conformal Anchors. It shows that, whenever the ordinary split-conformal threshold dominates the auxiliary perturbation threshold with high probability, the method can use $\hat{q}$ directly while retaining meaningful marginal guarantees.

B.4 Proof of Theorem 1.

Theorem 1 Assume (A1) – (A7). Let $\eta = \exp(-2n\gamma_F^2) + \exp(-2n\gamma_G^2)$. Then using the ordinary split-conformal threshold $\hat{q}$ yields:

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1) \leq \alpha + \delta + \eta.$$

Proof. Let

$$D_\rho = \{\tilde{q}_\rho \leq \hat{q}\}$$

be the threshold-dominance event. By Lemma 3,

$$\mathbb{P}(D_\rho^c) \leq \eta.$$

By Lemma 2, the probability of returning a Conformal Anchor for the true label is bounded by:

$$\alpha + \delta.$$

We decompose

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1) = \mathbb{P}(A_Y(\mathbf{X}_O) = 1, D_\rho) + \mathbb{P}(A_Y(\mathbf{X}_O) = 1, D_\rho^c).$$

The second term is bounded as

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1, D_\rho^c) \leq \mathbb{P}(D_\rho^c) \leq \eta.$$

On D_ρ , the threshold used by the method dominates the auxiliary perturbation-calibrated threshold. Hence, by the previous lemma,

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1, D_\rho) \leq \alpha + \delta.$$

Therefore,

$$\mathbb{P}(A_Y(\mathbf{X}_O) = 1) \leq \alpha + \delta + \eta$$

q.e.d.

In summary, perturbation calibration is used only as an analytical device to characterize the requirements for safe anchor-based exclusions. When the ordinary split-conformal threshold dominates, Conformal Anchors use it to obtain a marginal guarantee against returning an anchor that excludes the true label. Because this dominance event is not enforced directly, the bound includes the term η . Although η is not evaluated explicitly, it decays exponentially in the calibration set size whenever the separation margins γ_F, γ_G are non-negligible. Under this assumption and for practically sized calibration sets η is then expected to be small. The final guarantee therefore decomposes into three components: conformal error α , anchor-certification error δ , and threshold-dominance error η . We conclude the analysis providing in the next Corollary a conservative control over erroneous rejections when multiple anchors are returned for the same hypothesis.

B.5 Proof of Corollary 1.

Corollary 1 (Budgeted control for multiple returned anchors). Assume (A1) – (A7) and to return at most C anchors $A_Y^{(1)}, \dots, A_Y^{(C)}$ for excluding any label. For each c , suppose the guarantee of Theorem 1 holds with budgets:

$$\alpha' := \frac{\alpha}{C}, \quad \delta' := \frac{\delta}{C}, \quad \eta' := \frac{\eta}{C}$$

then

$$\mathbb{P}\left(\exists c \in \{1, \dots, C\} : A_Y^{(c)}(\mathbf{X}_O) = 1\right) \leq \alpha + \delta + \eta.$$

Proof. By the union bound,

$$\mathbb{P}\left(\exists c : A_Y^{(c)}(\mathbf{X}_O) = 1\right) = \mathbb{P}\left(\bigcup_{c=1}^C \{A_Y^{(c)}(\mathbf{X}_O) = 1\}\right) \leq \sum_{c=1}^C \mathbb{P}\left(A_Y^{(c)}(\mathbf{X}_O) = 1\right).$$

Using the single-anchor guarantee for each c ,

$$\sum_{c=1}^C \mathbb{P}\left(A_Y^{(c)}(\mathbf{X}_O) = 1\right) \leq \sum_{c=1}^C (\alpha' + \delta' + \eta').$$

By the budget constraints,

$$\sum_{c=1}^C (\alpha' + \delta' + \eta') \leq \alpha + \delta + \eta.$$

Then we conclude,

$$\mathbb{P}\left(\exists c : A_Y^{(c)}(\mathbf{X}_O) = 1\right) \leq \alpha + \delta + \eta.$$

q.e.d.

C Synthetic Dataset Construction

We use a synthetic medical diagnosis dataset to evaluate Conformal Anchors in a controlled partial-information setting. The goal is not to reproduce a real clinical population, but to create an interpretable multiclass problem with overlapping disease evidence, noisy measurements, and informative missingness.

Disease classes and features. Each instance is a synthetic patient with one of ten disease labels, sampled uniformly. The labels are divided into two anatomical groups. Lung-related diseases, consisting of asthma, bronchiectasis, bronchitis, COPD, and pneumonia. Stomach-related diseases, consisting of atrophic gastritis, GERD, gastric cancer, gastritis, and peptic ulcer. This structure induces ambiguity within each group, while making the two groups more separable.

Each patient is described by general symptoms and diagnostic tests. Symptoms are shared across all diseases and include fever, cough, chest pain, abdominal pain, fatigue, and nausea, each measured on a 0–10 severity scale. Diagnostic tests are group-specific. Lung-related tests include FEV1% predicted, chest X-ray severity, sputum neutrophil percentage, and white blood cell count. Stomach-related tests include endoscopy severity, *H. pylori* level, hemoglobin concentration, and gastric pH.

Data generation. For each patient, we first sample a disease label and then generate symptoms and test values conditionally on that label. Symptoms are drawn from disease-specific Beta distributions and rescaled to $[0, 10]$. These distributions encode broad group-level patterns, with respiratory symptoms more likely for lung diseases and gastrointestinal symptoms more likely for stomach diseases, while preserving overlap across diseases.

Diagnostic tests are generated from disease-specific probability tables over clinically meaningful value bands. For ordinal tests, a band is sampled directly and a value is drawn from that band. For continuous tests, the band probabilities are converted into disease-specific means and variances using the law of total expectation and variance. Then, values are sampled from truncated normal distributions within plausible physiological bounds. This produces continuous measurements while retaining interpretable disease-specific ranges.

Missingness and noise. Partial information is introduced through both structural and symptom-level missingness. Diagnostic tests are usually observed only for the anatomical system corresponding to the true disease group. Thus, tests from the other system are missing, except with small probability (0.5% for each out-of-group test), in which case they are sampled from healthy reference ranges. Symptoms may also be missing in a group-dependent way: respiratory symptoms are more often observed for lung diseases, while gastrointestinal symptoms are more often observed for stomach diseases. Fatigue is treated as non-specific and has intermediate missingness.

Finally, clinical variability is introduced through probabilistic sampling and additive Gaussian noise with mean 0 and standard deviation 0.5 on symptom scores, followed by clipping to the valid range $[0, 10]$. As a result, uncertainty arises from overlapping disease profiles, noisy observations, and missing information rather than from class imbalance.

C.1 Disease-specific diagnostic test distributions

The probabilistic generation of diagnostic-test values relies on disease-specific value-band tables. These tables provide the numerical specifications underlying the diagnostic-test generation mechanism described above. For each disease and each associated diagnostic test, probability mass is assigned to clinically plausible value ranges. The resulting distributions encode characteristic, but non-deterministic, diagnostic profiles for each condition.

Tables induce structured diagnostic ambiguity: different diseases may place high probability on different value ranges, but the distributions are deliberately overlapping so no diagnostic test deterministically identifies a unique disease.

For ordinal diagnostic tests, the value bands are used directly as categorical distributions over possible score ranges. For continuous diagnostic tests the same tables are used to derive disease-specific sampling distributions. Then, each value band is treated as a uniform distribution over its interval and band-level moments are combined according to the assigned probabilities. This yields disease-specific mean and variance used for truncated-normal sampling.

Finally, we report in Table 15 the healthy reference ranges used when, with low probability (0.5% for each out-of-group diagnostic test), a diagnostic test belonging to the non-affected anatomical system is observed. We report the value-band tables separately for lung-related (Tables 5–9) and stomach-related (Tables 10–14) diseases.

Lung-related diseases

Table 5: Value bands and probability masses for *bronchitis* diagnostic tests.

Diagnostic test	Value range	Probability
Pulmonary function (%)	70–100	0.75
	60–69	0.15
	50–59	0.06
	40–49	0.02
	30–39	0.01
	15–29	0.01
Chest X-ray score	0	0.70
	1–2	0.20
	3–4	0.07
	5–7	0.02
	8–10	0.01
Sputum neutrophils (%)	20–39	0.02
	40–54	0.08
	55–69	0.30
	70–84	0.45
	85–100	0.15
WBC count (cells/ μ L)	3000–6000	0.40
	6001–9000	0.40
	9001–12000	0.12
	12001–16000	0.06
	16001–20000	0.01
	20001–25000	0.01

Table 6: Value bands and probability masses for *COPD* diagnostic tests.

Diagnostic test	Value range	Probability
Pulmonary function (%)	70–100	0.01
	60–69	0.03
	50–59	0.06
	40–49	0.20
	30–39	0.30
	15–29	0.40
Chest X-ray score	0	0.02
	1–2	0.08
	3–4	0.30
	5–7	0.40
	8–10	0.20
Sputum neutrophils (%)	20–39	0.30
	40–54	0.40
	55–69	0.20
	70–84	0.07
	85–100	0.03
WBC count (cells/ μ L)	3000–6000	0.30
	6001–9000	0.45
	9001–12000	0.18
	12001–16000	0.05
	16001–20000	0.01
	20001–25000	0.01

Table 7: Value bands and probability masses for *pneumonia* diagnostic tests.

Diagnostic test	Value range	Probability
Pulmonary function (%)	70–100	0.02
	60–69	0.05
	50–59	0.10
	40–49	0.20
	30–39	0.28
	15–29	0.35
Chest X-ray score	0	0.00
	1–2	0.01
	3–4	0.09
	5–7	0.40
	8–10	0.50
Sputum neutrophils (%)	20–39	0.00

Diagnostic test	Value range	Probability
	40–54	0.02
	55–69	0.10
	70–84	0.38
	85–100	0.50
WBC count (cells/ μ L)	3000–6000	0.01
	6001–9000	0.03
	9001–12000	0.10
	12001–16000	0.26
	16001–20000	0.30
	20001–25000	0.30

Table 8: Value bands and probability masses for *asthma* diagnostic tests.

Diagnostic test	Value range	Probability
Pulmonary function (%)	70–100	0.78
	60–69	0.15
	50–59	0.04
	40–49	0.02
	30–39	0.005
	15–29	0.005
Chest X-ray score	0	0.80
	1–2	0.15
	3–4	0.04
	5–7	0.005
	8–10	0.005
Sputum neutrophils (%)	20–39	0.50
	40–54	0.30
	55–69	0.12
	70–84	0.06
	85–100	0.02
WBC count (cells/ μ L)	3000–6000	0.30
	6001–9000	0.50
	9001–12000	0.12
	12001–16000	0.06
	16001–20000	0.01
	20001–25000	0.01

Table 9: Value bands and probability masses for *bronchiectasis* diagnostic tests.

Diagnostic test	Value range	Probability
Pulmonary function (%)	70–100	0.05
	60–69	0.10
	50–59	0.20
	40–49	0.30
	30–39	0.20
	15–29	0.15
Chest X-ray score	0	0.001
	1–2	0.029
	3–4	0.20
	5–7	0.45
	8–10	0.32
Sputum neutrophils (%)	20–39	0.001
	40–54	0.039
	55–69	0.18
	70–84	0.38
	85–100	0.40
WBC count (cells/ μ L)	3000–6000	0.05
	6001–9000	0.15
	9001–12000	0.35
	12001–16000	0.30
	16001–20000	0.10
	20001–25000	0.05

Stomach-related diseasesTable 10: Value bands and probability masses for *gastritis* diagnostic tests.

Diagnostic test	Value range	Probability
Endoscopy score	0	0.25
	1–2	0.45
	3–4	0.20
	5–6	0.06
	7–8	0.03
	9–10	0.01
<i>H. pylori</i> level	0	0.02
	1	0.08
	2	0.30
	3	0.60
Hemoglobin (g/dL)	5–8	0.01

Diagnostic test	Value range	Probability
	8.1–10	0.02
	10.1–12	0.10
	12.1–14	0.45
	14.1–16	0.30
	16.1–18	0.12
Gastric pH	1–2	0.70
	3	0.18
	4	0.06
	5	0.03
	6–7	0.02
	8	0.01

Table 11: Value bands and probability masses for *gastric cancer* diagnostic tests.

Diagnostic test	Value range	Probability
Endoscopy score	0	0.00
	1–2	0.00
	3–4	0.02
	5–6	0.08
	7–8	0.30
	9–10	0.60
<i>H. pylori</i> level	0	0.25
	1	0.25
	2	0.25
	3	0.25
Hemoglobin (g/dL)	5–8	0.08
	8.1–10	0.12
	10.1–12	0.30
	12.1–14	0.30
	14.1–16	0.15
	16.1–18	0.05
Gastric pH	1–2	0.05
	3	0.10
	4	0.15
	5	0.20
	6–7	0.25
	8	0.25

Table 12: Value bands and probability masses for *peptic ulcers* diagnostic tests.

Diagnostic test	Value range	Probability
Endoscopy score	0	0.02
	1–2	0.05
	3–4	0.18
	5–6	0.45
	7–8	0.25
	9–10	0.05
<i>H. pylori</i> level	0	0.05
	1	0.10
	2	0.30
	3	0.55
Hemoglobin (g/dL)	5–8	0.05
	8.1–10	0.10
	10.1–12	0.25
	12.1–14	0.35
	14.1–16	0.20
	16.1–18	0.05
Gastric pH	1–2	0.75
	3	0.15
	4	0.06
	5	0.02
	6–7	0.01
	8	0.01

Table 13: Value bands and probability masses for *atrophic gastritis* diagnostic tests.

Diagnostic test	Value range	Probability
Endoscopy score	0	0.05
	1–2	0.25
	3–4	0.40
	5–6	0.20
	7–8	0.07
	9–10	0.03
<i>H. pylori</i> level	0	0.70
	1	0.20
	2	0.07
	3	0.03
Hemoglobin (g/dL)	5–8	0.06

Diagnostic test	Value range	Probability
	8.1–10	0.12
	10.1–12	0.30
	12.1–14	0.34
	14.1–16	0.12
	16.1–18	0.06
Gastric pH	1–2	0.02
	3	0.05
	4	0.08
	5	0.15
	6–7	0.40
	8	0.30

Table 14: Value bands and probability masses for *GERD* diagnostic tests.

Diagnostic test	Value range	Probability
Endoscopy score	0	0.55
	1–2	0.35
	3–4	0.07
	5–6	0.02
	7–8	0.005
	9–10	0.005
<i>H. pylori</i> level	0	0.80
	1	0.12
	2	0.06
	3	0.02
Hemoglobin (g/dL)	5–8	0.01
	8.1–10	0.02
	10.1–12	0.05
	12.1–14	0.50
	14.1–16	0.30
	16.1–18	0.12
Gastric pH	1–2	0.70
	3	0.18
	4	0.06
	5	0.03
	6–7	0.02
	8	0.01

Table 15: Healthy reference ranges for diagnostic tests used in the synthetic dataset.

Diagnostic test	Healthy range
Pulmonary function (%)	85–100
Sputum neutrophils (%)	10–50
WBC count (cells/ μ L)	4500–9000
Hemoglobin (g/dL)	12.5–16.5
Gastric pH	1.5–3.5
Chest X-ray score	0–1
Endoscopy score	0–1
<i>H. pylori</i> level	0–1

C.2 Healthy reference ranges

Table 15 reports the reference intervals used to generate physiologically normal values for out-of-group diagnostic tests. These values are used only with low-probability to simulate unnecessary, broad, or procedurally noisy testing.

C.3 Samples

We report in Table 16 some full random samples from both the training and test distribution.

D Implementation Details

D.1 Model training and calibration

Training. The predictive model adopted in this work is an XGBoost multiclass classifier. The model uses a softmax prediction head, which directly produces class-conditional probability estimates for each patient instance. Missing values are not imputed, since they are part of the data-generating process and carry information about which clinical evidence is available. Moreover, missing values are natively handled by XGBoost, which learns default directions for missing values at tree splits. No feature scaling is applied, because the predictive model is tree-based and the original feature units are clinically interpretable. Hyperparameters (detailed in Table 17) are selected monitoring training set performance.

Calibration. We leverage standard split-conformal calibration with a target $1 - \alpha$ guarantee set to 0.95. The resulting conformal threshold is $\hat{q} = 0.9687$, meaning that all instances with assigned confidence greater than 0.0313 were kept in the conformal prediction set. We detail in Fig. 3 the procedure to construct the conformal prediction set for a random sample.

We report in Table 18, accuracy and average conformal prediction set sizes for both training and test sets.

Table 16: Two representative synthetic patients, one for each disease group, and two test-time patient, one for each disease group, illustrating the dataset structure under partial information.

Feature	Patient 1 (Pneumonia)	Patient 2 (Gastritis)	Test-time patient 1 (Peptic Ulcer)	Test-time patient 2 (COPD)
<i>General symptoms (0-10 scale)</i>				
Fever severity	7	3	3	4
Cough severity	7	0	4	8
Chest pain severity	4	0	2	3
Abdominal pain severity	0	9	9	3
Fatigue level	3	1	6	5
Nausea	2	3	7	4
<i>Lung-related diagnostic tests</i>				
FEV1% predicted	34.5	—	—	—
Chest X-ray score (0-10)	9	—	—	—
Sputum neutrophils (%)	77.0	—	—	—
WBC count (cells/ μ L)	18,000.5	—	—	—
<i>Gastric diagnostic tests</i>				
Endoscopy score (0-10)	—	1.5	—	—
H. pylori level (0-3)	—	3	—	—
Hemoglobin (g/dL)	—	15.05	—	—
Gastric pH	—	1.5	—	—

Table 17: Hyper-parameters of the XGBoost classifier used in the experiments.

n_estimators	max_depth	learning_rate	min_child_weight	subsample	gamma	reg_alpha	reg_lambda
200	5	0.03	6	0.7	0.2	0.3	0.1

D.2 Anchor construction

Preprocessing. To extract explanations, we discretize continuous variables into intervals. Since anchor are expressed as logical rules, feature predicates based on clinically meaningful intervals are more interpretable than predicates involving numerical thresholds. Generic symptoms are discretized into integer severity levels from 0 to 10. This is obtained by partitioning the original $[0, 10]$ scale into unit-width intervals and assigning each observed value to the corresponding severity category. Diagnostic tests are discretized using the same value bands used in the generative process. Continuous test are mapped to the midpoint of the band in which they fall. Ordinal tests are mapped to the corresponding ordinal category or representative band value. This ensures consistency between the data-generating mechanism and the rules returned by Conformal Anchors.

Hyperparameters. We report all hyperparameters used for the anchor search in Table 19. We set $\tau = 0.95$ and $\delta = 0.01$, requiring anchors to satisfy a high-

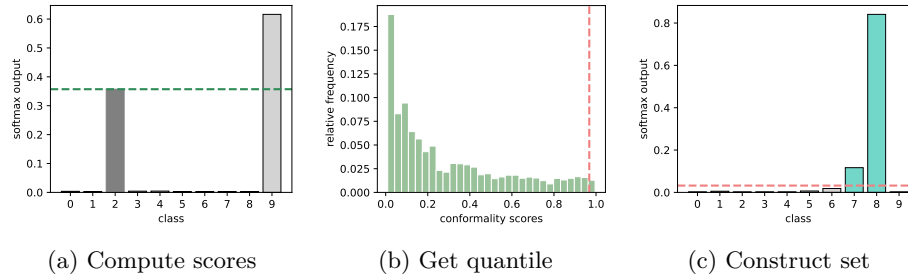


Fig. 3: Illustration of our split conformal prediction implementation. The calibration scores are computed on held-out data (Fig. 3a), the empirical quantile $\hat{q}$ is extracted from their distribution (Fig. 3b), and the prediction set for a test instance is obtained by thresholding the softmax outputs at $1 - \hat{q}$ (Fig. 3c).

Table 18: Accuracy and average prediction set size for training and test set.

Split	Accuracy	Avg. conformal prediction set size
Training	0.8120	2.106
Test	0.8085	2.097

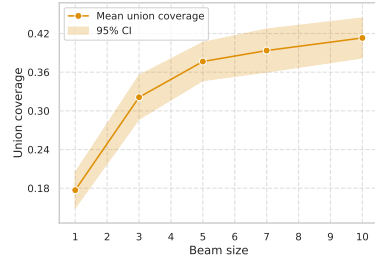
precision constraint with high confidence, as appropriate in high-stakes decision-support settings. The union pruning threshold is set to 10^{-4} , allowing the search to retain even low-frequency candidate regions in the neighborhood, which is particularly important for the medoid-based strategies. Finally, we select a beam size of $B = 5$ based on the trade-off between search exhaustiveness and computational cost. To choose this value, we use the anchor pool to evaluate, for each instance and for different values of B , the union coverage of the set of returned anchors for a random query. We then compare the mean union coverage with the corresponding runtime. As shown in (Fig. 4), union coverage saturates around $B = 5$ for all methods, suggesting that larger beam sizes increase runtime while providing negligible gains in search exhaustiveness.

E Hardware

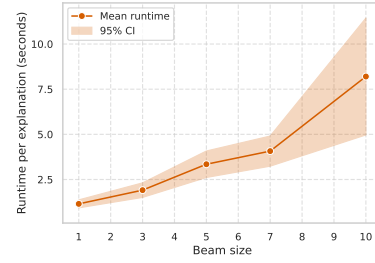
For our experiments, we use a 16-core machine with an AMD Ryzen 9 7950X CPU and 2 NVIDIA GeForce RTX 4090 GDDR6X with 24GB of memory, OS Ubuntu 22.04.4 LTS.

Table 19: Hyperparameters for Conformal Anchor search.

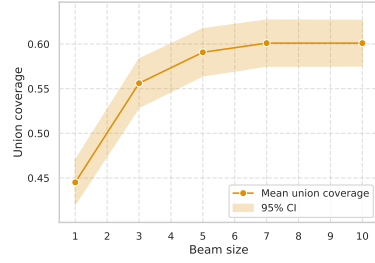
Hyperparameter	Value	Description
τ	0.95	Minimum precision threshold
δ	0.01	Precision certification tolerance
ε	10^{-4}	Union pruning threshold
B	5	Beam size



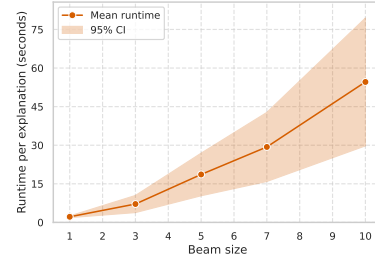
(a) OS: Beam size vs. union coverage



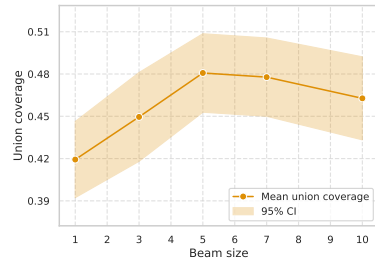
(b) OS: Beam size vs. runtime



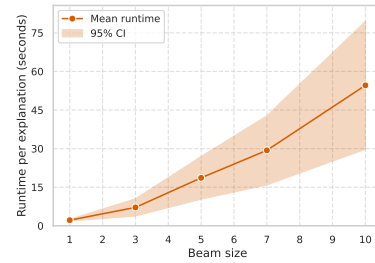
(c) MG: Beam size vs. union coverage



(d) MG: Beam size vs. runtime



(e) CG: Beam size vs. union coverage



(f) CG: Beam size vs. runtime

Fig. 4: Beam-size selection analysis for the three construction modes. The left column reports union coverage as a function of the beam size B , while the right column reports the corresponding runtime. Coverage tends to saturate around $B = 5$, whereas runtime continues to increase, motivating our choice of $B = 5$.